

Assignment-2

Date:

Page:

1. State and prove De-Morgan's Theorem with necessary

⇒

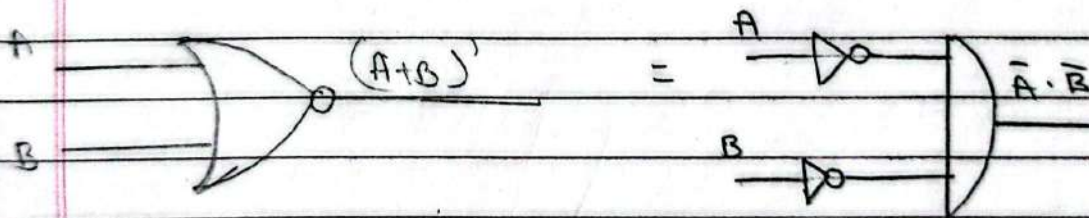
Theorem-1

The first De-Morgan's Theorem states that the complement of a sum is the product of the complements.

$$\text{i.e. } (A+B)' = A' \cdot B'$$

Truth-Table

Input		Output				
A	B	A+B	$(A+B)'$	A'	B'	$A' \cdot B'$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0



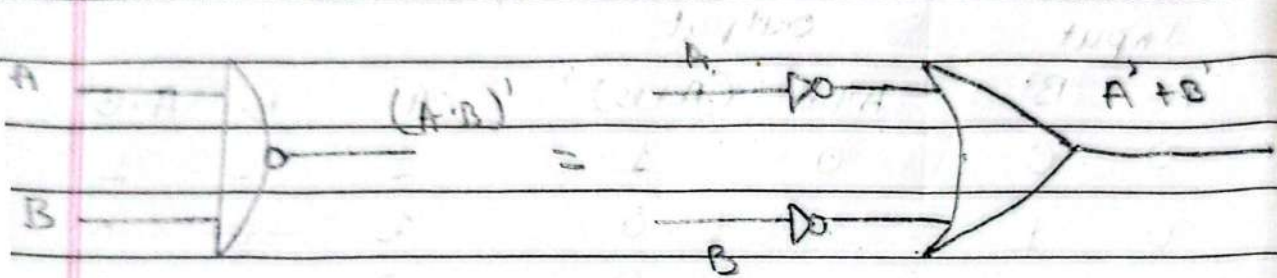
Theorem-2

The complement of a product is the sum of the complements.

$$\text{i.e. } (A \cdot B)' = A' + B'$$

Truth Table

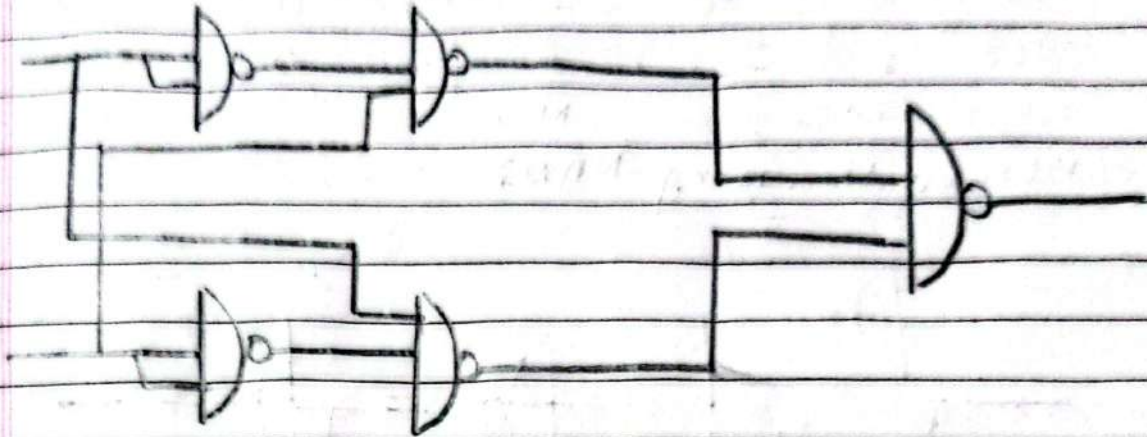
A	B	$A \cdot B$	$(A \cdot B)'$	A'	B'	$A' + B'$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0



2. Realise Ex-OR, Ex-NOR, NOR gates using NAND gates only,

⇒

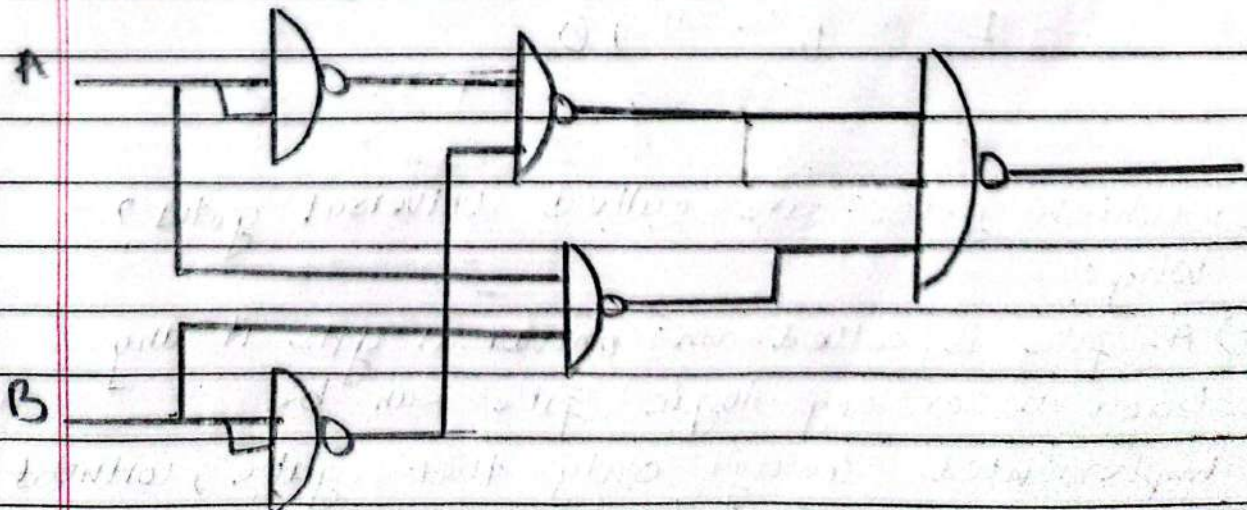
a. Ex-OR gate using NAND gates.



Truth-Table

Input		Output				
A	B	$\bar{A}$	$A'B$	$B'\bar{A}$	AB'	$A'B + AB'$
0	0	1	0	1	0	0
0	1	1	1	0	0	1
1	0	0	0	1	1	1
1	1	0	0	0	0	0

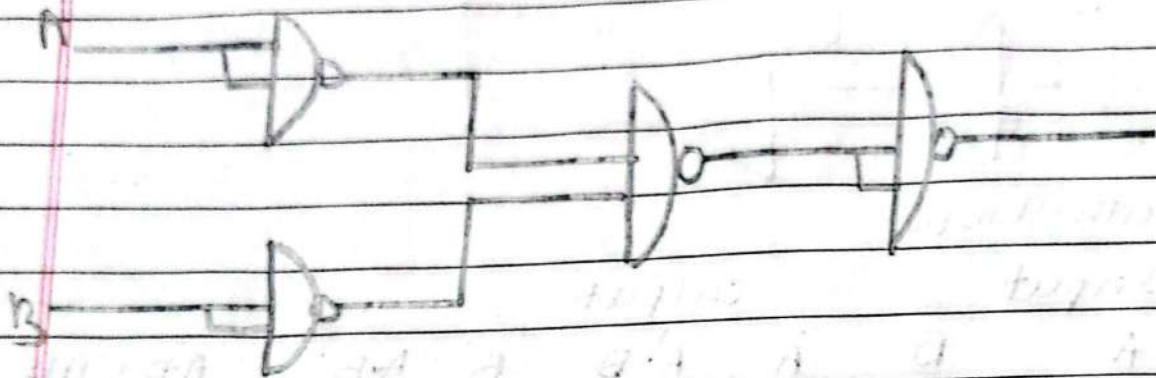
b. Ex-NOR gate using NAND.



Truth-Table

Input		Output				
A	B	A'	B'	$A \cdot B$	$A' \cdot B'$	$A \cdot B + A' \cdot B'$
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1

c. NOR gate using NAND



Truth-Table

Input		Output	
A	B	$A + B$	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

3. Which gates are called universal gates?
Why?

→ A gate is called an universal gate if any boolean or any logic gate can be implemented using only that gate, without

needing any other type of gate. NAND and NOR gates are considered universal gates because NOT, AND, OR, the three basic gates can all be built from either NAND or NOR gate alone.

4. Mention the steps to get the Dual of any Boolean expression with example.

=> The dual of a Boolean expression is obtained by replacing every AND with OR, OR with AND and every 0 with 1 and vice-versa, while keeping the variables unchanged.

Eg:

$$1) A + 0 = A \cdot 1 = A$$

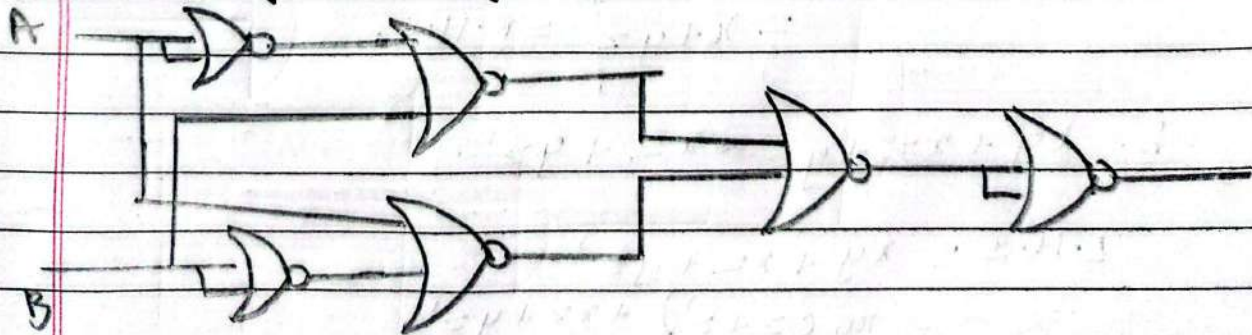
$$2) A(B + C) = A \cdot B + A \cdot C$$

$$= (A + B) \cdot (A + C)$$

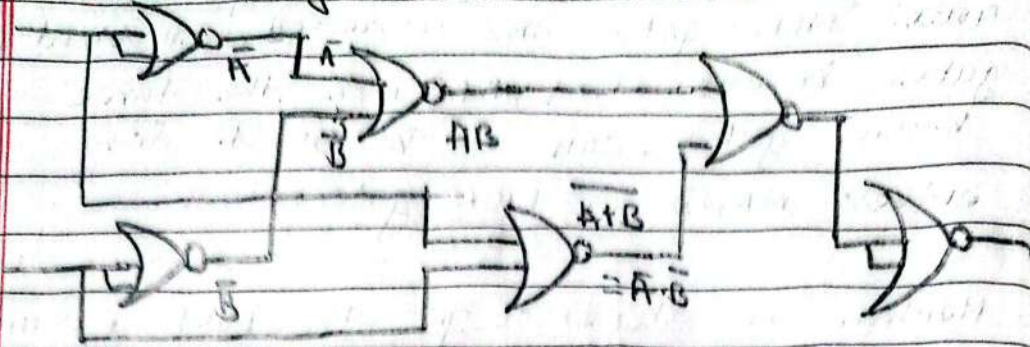
5. Realize EX-OR, EX-NOR, NAND gates using NOR gates only.

=>

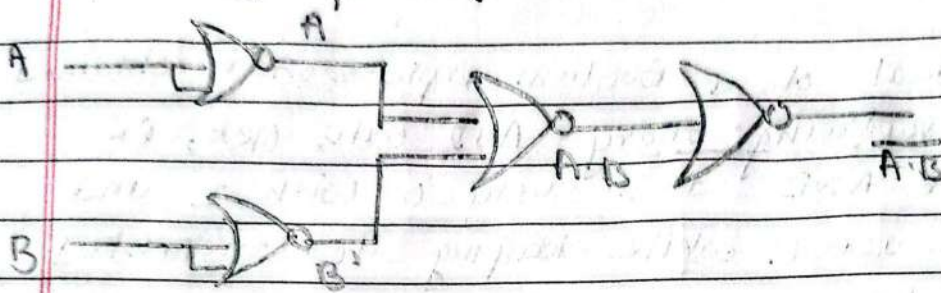
Ex-OR gate using NOR $Y = A'B + AB'$



X-NOR using NOR $Y = A'B + \overline{A}B$



NAND gate using NOR



6. Prove that:

a. $(x+y)(x+z) = x + yz$

L.H.S = $(x+y)(x+z)$

$$= x \cdot x + x \cdot z + y \cdot x + y \cdot z$$

$$= x(1+z) + xy + yz$$

$$= x \cdot 1 + xz + yz$$

$$= x(1+y) + yz$$

$$= x \cdot 1 + yz$$

$$= x + yz = R.H.S.$$

b. $xy + xz + yz = xz + yz'$

L.H.S = $xy + xz + yz'$

$$= xy(z+z') + xz + yz'$$

$$= xyz + xy z' + xz + yz'$$

$$= xyz + xz + xy z' + yz'$$

$$\begin{aligned}
 &= xz(y+1) + yz'(x+1) \\
 &= xz + yz' \\
 &= \text{R.H.S. proved}
 \end{aligned}$$

c. $x(x+y) = x$

$$\begin{aligned}
 \text{L.H.S.} &= x(x+y) \\
 &= xx + xy \\
 &= x + xy \\
 &= x(1+y) \\
 &= x = \text{R.H.S. proved}
 \end{aligned}$$

d. $x + xy = x$

$$\begin{aligned}
 \text{L.H.S.} &= x + xy \\
 &= x\left(\frac{1}{x} + y\right) \\
 &= x = \text{R.H.S. proved}
 \end{aligned}$$

e. $A + B'C(A + B'C) = A + B'C$

$$\begin{aligned}
 \text{L.H.S.} &= A + B'C(A + B'C) \\
 &= A + AB'C + B'B'C \\
 &= A + AB'C + B'C \\
 &= A + B'C(1 + A) \\
 &= A + B'C \\
 &= \text{R.H.S. proved}
 \end{aligned}$$

f. $AB + A(B+C) + B(B+C) = B + AC$

$$\begin{aligned}
 \text{L.H.S.} &= AB + A(B+C) + B(B+C) \\
 &= AB + AB + AC + B.B + B.C \\
 &= AB + AC + B + BC \\
 &= B(A+1) + C(A+B) \\
 &= B + BC + AC \\
 &= B(1+C) + AC = B + AC = \text{R.H.S.}
 \end{aligned}$$

Assignment - 3,

$$1. F = \sum m(0, 2, 4, 6, 11, 13)$$

$$d = (8, 10, 12, 14)$$

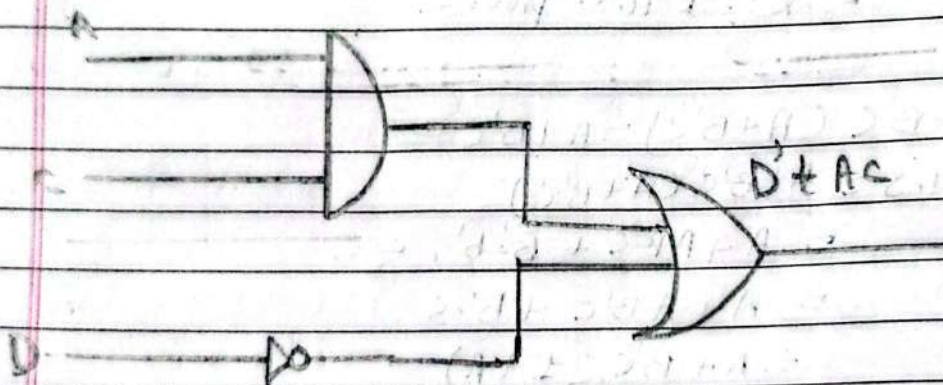
a. basic gates only.

b. NAND gates only

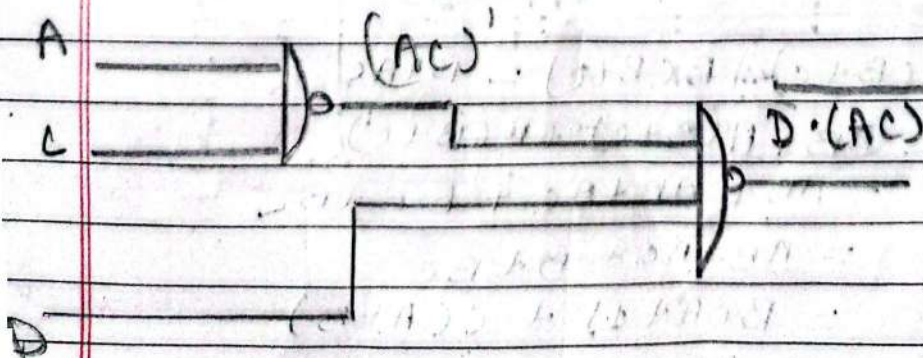
AB\CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	1	X
10	X	0	1	X

$$F = D' + AC = \overline{(D)} + (AC) = \overline{(D \cdot (AC)')}$$

a) Using basic gate



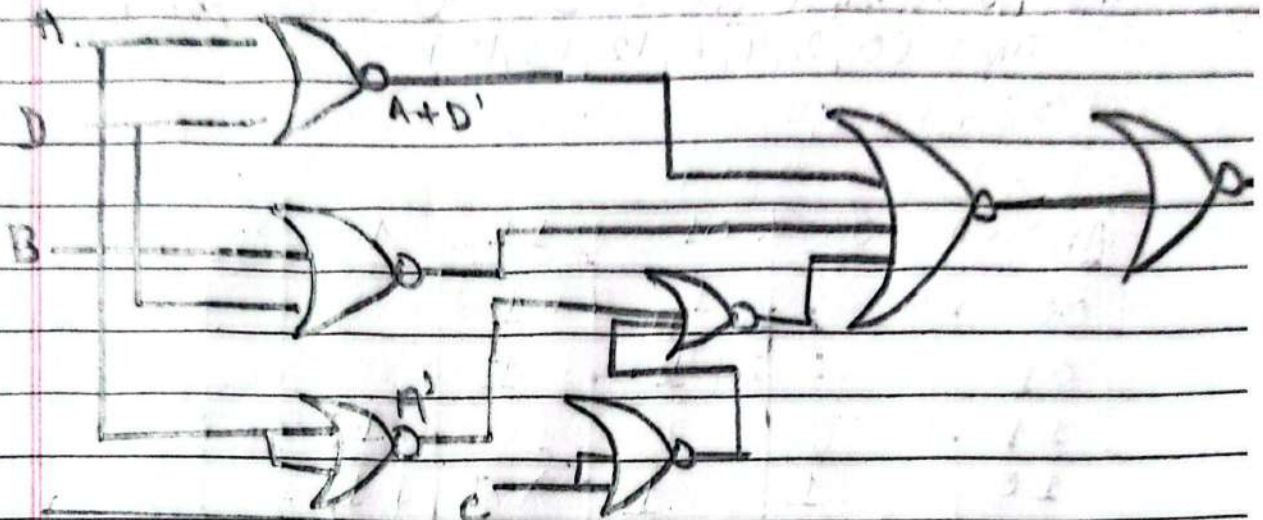
b) Using NAND gate only



2. $F(A, B, C, D) = \sum m(0, 2, 4, 6, 8, 10, 15) + d(3, 11, 14)$
 a) Represent using NOR gate.

AB \ CD	00	01	11	10
00	1		X	1
01	1		0	1
11	0		1	X
10	1		X	1

$$F = A'D' + B'D' + AC = (A+D) \cdot (B+D) \cdot (A'+A'C')$$



3. $\pi_m = (0, 4, 5, 8, 11, 15)$

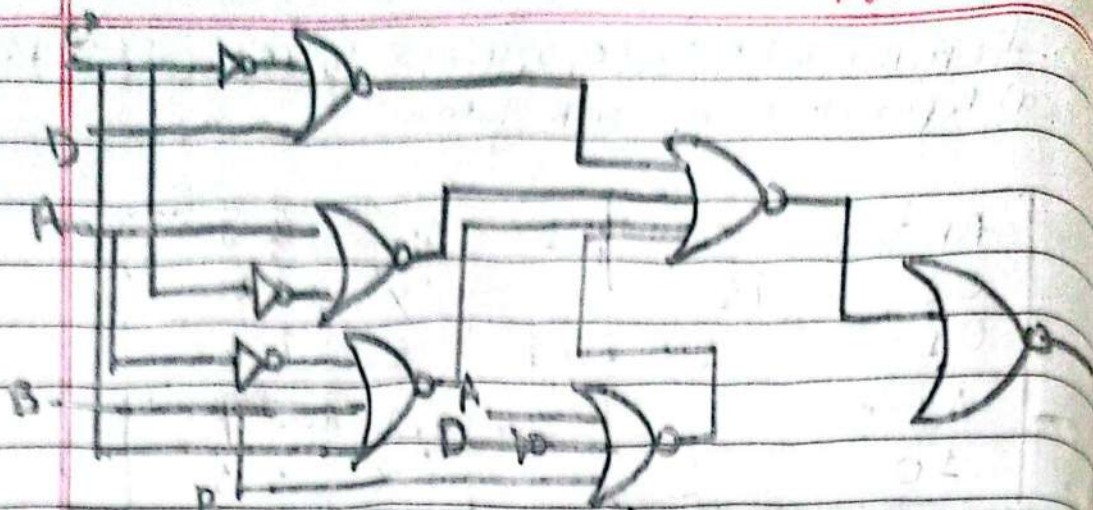
AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	1	1
11	1	1	0	1
10	0	0	0	1

$$F = (A+C+D) (A+B'+C) (A'+C'+D) (A'+B+C)$$

$$F = \pi_m(0, 4, 5, 8, 11, 15)$$

$$F = \sum m(1, 2, 3, 6, 7, 9, 10, 12, 13, 14)$$

$$F = (C'D) + (A+C') + (A'B+C) + (A+B'D')$$



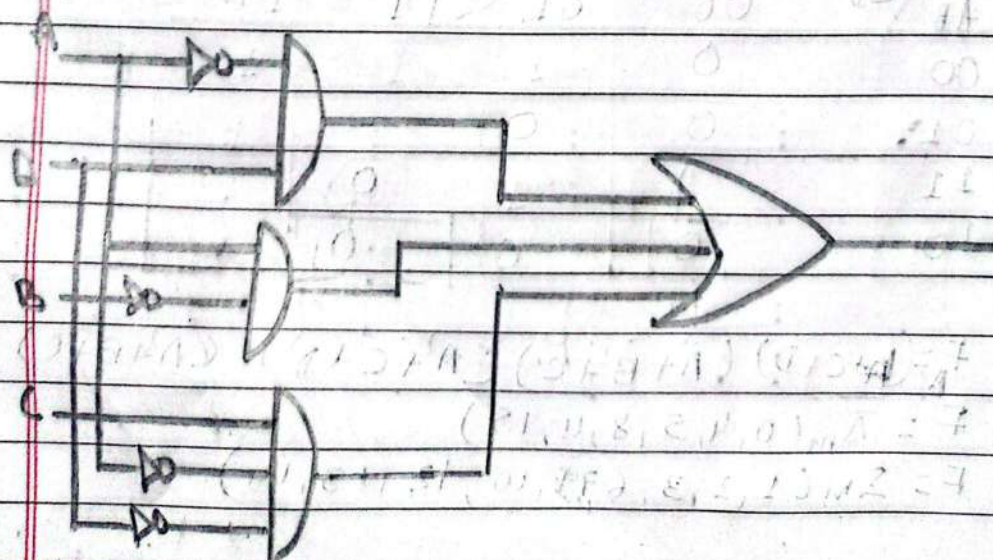
4. $\Sigma_m = (1, 3, 5, 7, 8, 9, 10, 11, 14)$

For POS,

$$\pi_m = (0, 2, 4, 6, 12, 13, 15)$$

AB \ CD	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	0	0	1
10	1	1	1	1

$$F = \Sigma_m (A'D + AB' + A'CD')$$

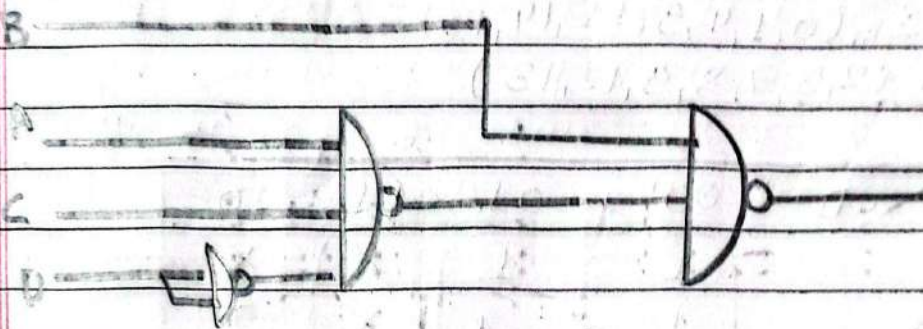


5. $F = \sum m(1, 2, 3, 8, 9, 10, 11, 14)$

$D = (0, 4, 12)$

AB \ CD	00	01	11	10
00	X	1	1	1
01	X	0	0	0
11	X	0	0	1
10	1	1	1	1

$$F = B' + ACD' = (B' + ACD')' = (C(B')' + (ACD')')' = B \cdot ACD'$$



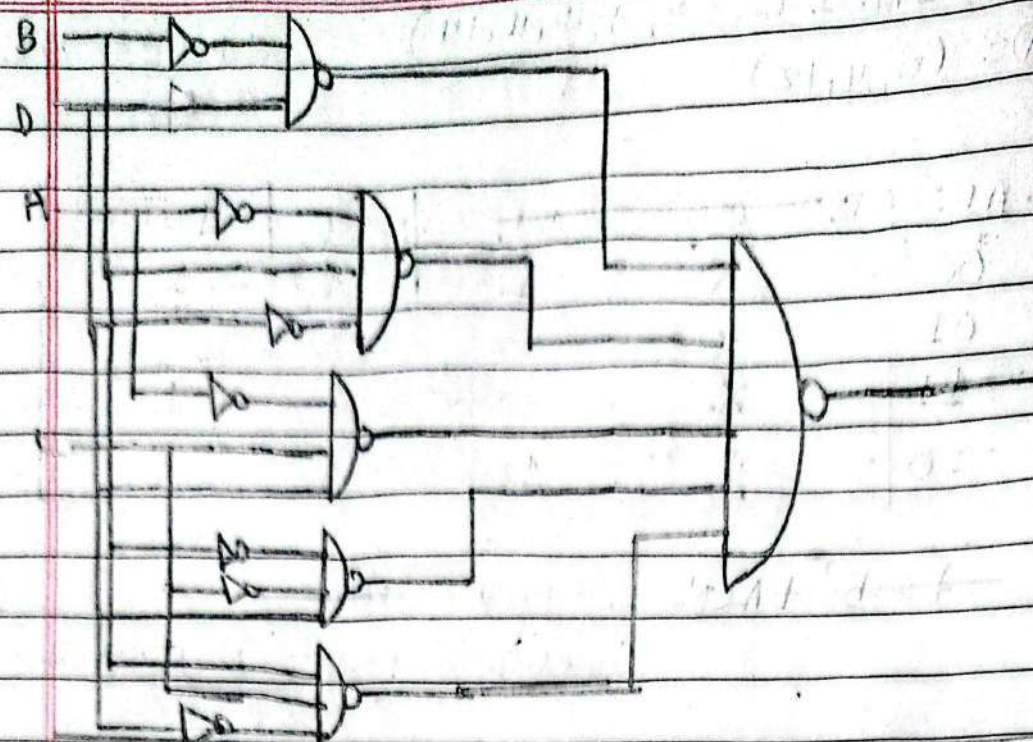
6. $F = \sum m(3, 4, 7, 8, 14)$

$D = (1, 6, 9, 11, 13)$

AB \ CD	00	01	11	10
00	0	X	1	0
01	1	0	1	X
11	0	X	0	1
10	1	X	X	0

$$F = B'D + A'BD' + AB'C' + B'CD' + A'CD$$

$$= ((B'D + A'BD' + AB'C' + B'CD' + A'CD)')$$



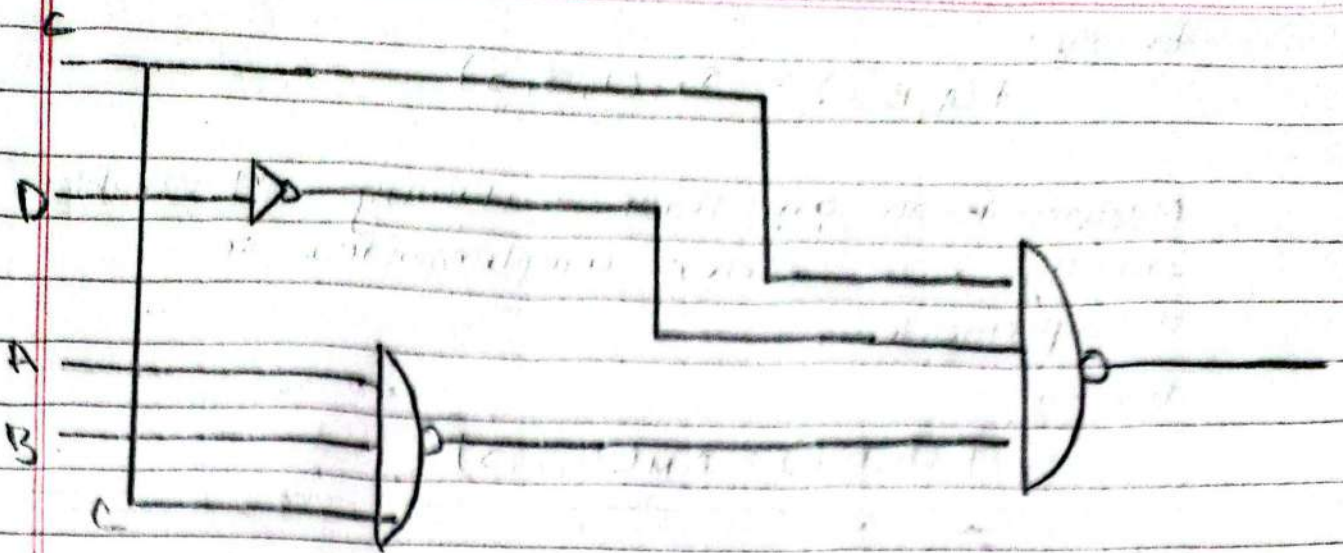
$$F = \sum m(0, 1, 4, 5, 11, 14, 13)$$

$$D = (2, 3, 7, 8, 9, 12, 13)$$

AB \ CD	00	01	11	10
00	1	1	X	X
01	1	1	X	0
11	X	X	1	1
10	X	X	1	0

$$F = C' + CD + ABC$$

$$= C \cdot D' \cdot ABC$$



8. Define SOP and POS with example.

⇒ SOP is a Boolean expression written as the OR (+) of two or more AND (·) terms.

Eg:

$$F(A, B, C) = AB' + A'C + BC$$

Product of Sum is a Boolean written as the AND (·) of two or more OR (+) terms.

Eg:

$$F = (A+B)(\bar{A}+C)(B+C)$$

9. Define Minterm and Maxterms with example.

⇒ Minterm is an AND term containing all variables exactly once, either complemented or uncomplemented.

For eg:

$$F(A, B, C) = \sum m(1, 3, 5)$$

Maxterm is an OR term containing all variables exactly once, either complemented or uncomplemented,

For eg:

$$F(A, B, C) = \prod M(0, 2, 5)$$

Q. Define the principle of duality.

⇒ The principle of duality states that:

"If a Boolean identity is true, then another valid identity can be obtained by interchanging AND and OR operators and also interchanging 0 and 1."